Week 2: Casework and Advanced Methods

Problem 1

There are 5 Democrat and 3 Republican senators on a committee. How many ways are there to seat them in a row of 8 seats if none of the Republicans wish to sit next to each other?

Problem 2

Amy and Jake are playing a game with a fair coin with heads on one side and tails on the other. They repeatedly flip the coin until either the sequence HH or TT appears, which stops the game. If HH appears, then Amy will win; if TT appears, then Jake will win.

a) What is the probability that Amy will win the game?

b) If H is confirmed as the first flip, what is the probability that Amy wins the game?

Problem 3

Three distinct points are selected randomly among the vertices of a cube.

a) What is the probability that not all three points lie on the same face?

b) What is the probability that the three points can form the vertices of a scalene triangle?

Problem 4

A class trip of 10 students is making an evening stop at an Austrian hotel. They have booked five rooms, each of which can house two people. Suppose that people are assigned randomly to rooms. What is the probability that three students, Peter, Ned, and Flash, are in separate rooms?

Problem 5

Four teams, the Wizards, Dragons, Jaguars, and Astronauts, are competing for a basketball championship. The Wizard and Dragons each have a 75% chance of defeating the Jaguars or Astronauts, and have a 50% chance against each other. Similarly, the Jaguars and Astronauts have a 50% chance against each other. The teams are randomly assigned to two semifinal matchups, and the winners of each semifinal will play in the finals. What is the probability that the Wizards win the championship?

Problem 6

A Mars rover is positioned at $(0, 0)$ in the coordinate plane. For each step, it will move with equal probability either one unit upward, downward, leftward, or rightward. After 7 steps, what is the probability that the Rover is within a distance of $\sqrt{5}$ of the origin?

Problem 7

Each person in a cohort of 20 recruits is randomly assigned to one of five factions: Abnegation, Amity, Candor, Dauntless, and Erudite. Each faction must receive at least one new recruit, and no more than 10 new recruits. How many possible distributions of new recruits to factions are possible, assuming that two distributions are considered indistinguishable if they provide the same number of new recruits to each faction?

Problem 8

Each of the positive integer divisors of 15! is written onto a pool ball. All the balls are placed inside a mason jar and mixed. A ball is randomly drawn from the jar.

a) What is the probability that the number written on the ball is odd?

b) What is the probability that the number written on the ball is a perfect square?

Problem 9

An activity involves 200 people and their phone numbers, which are written onto cards and distributed randomly, with one card provided to each person. The first person starts by calling the number on his card, and the person who receives this call will proceed by calling the number on their card, and so forth. The activity stops when a person receives a call twice. What is the probability that all 200 people will be called during the activity?

Remark: This combinatorial structure is typically called a Hamiltonian cycle.

Problem 10

Magnus flips n fair coins and Kristoph flips $n + 1$ fair coins. What is the probability that Kristoph flips more heads than Magnus?

Week 2: Casework and Advanced Methods SOLUTIONS

Problem 1

There are 5 Democrat and 3 Republican senators on a committee. How many ways are there to seat them in a row of 8 seats if none of the Republicans wish to sit next to each other?

Answer: $\boxed{14400}$

Solution 1: Designate each Democrat as _ and each Republican as R. The possible seating structures are illustrated below, stratified based on the first and second appearance of R.

$$\begin{bmatrix} R_R_R___ & R_R__R__ & R_R___R_ & R_R___R \\ R__R_R__ & R__R__R_ & R__R___R & \\ R___R_R_ & R___R__R & & \\ R____R_R & & & \\ & & & \\ _R_R_R__ & _R_R__R_ & _R_R___R & \\ _R__R_R_ & _R__R__R & & \\ _R___R_R & & & \\ & & & \\ __R_R_R_ & __R_R__R & & \\ __R__R_R & & & \\ & & & \\ ___R_R_R & & & \end{bmatrix}$$

There are 20 of these in total, with 5! ways to distribute the Democrats and 3! ways to distribute the Republicans in each case. This manufactures a total of $20 * 5! * 3! = \boxed{14400}$ seating arrangements.

Solution 2: Here, we introduce a faster way to count seating structures. We note that we can simply count the number of ways to arrange 3 Ds and 3 Rs since for each arrangement, we can simply place one Democrat on each side of the middle Republican. For instance, $RRRDDD \to RDRDRDDD$. There are $\binom{6}{3} = 20$ arrangements, and $5! * 3!$ to distribute the 5 Democrats and 3 Republicans, producing a total of $20 * 5! * 3! = \boxed{14400}$ seatings.

Problem 2

Amy and Jake are playing a game with a fair coin with heads on one side and tails on the other. They repeatedly flip the coin until either the sequence HH or TT appears, which stops the game. If HH appears, then Amy will win; if TT appears, then Jake will win.

a) What is the probability that Amy will win the game?

b) If H is confirmed as the first flip, what is the probability that Amy wins the game?

Answer: (a) $\boxed{\frac{1}{2}}$ (b) $\boxed{\frac{2}{3}}$

Solution: (a) We manufacture casework based on the first flip of the coin.

- *Case 1: First flip is H.* Let p denote the probability of Amy winning. There is a $\frac{1}{2}$ chance of Amy winning next flip (HH). If the second flip is tails, the third flip must be heads to prevent a Jake win, which occurs with $\frac{1}{4}$ chance and resets the probabilities as we return to a most recent flip of H. Thus, $p = \frac{1}{2} + \frac{1}{4} * p$, so $p = \frac{2}{3}$.
- *Case 2: First flip is T.* Let p denote the chance of Amy winning. For Amy to win, the second flip must be heads (TH) with $\frac{1}{2}$ chance, and we revert to the situation from Case 1. Thus, the probability of Amy winning here is hence $\frac{1}{2} * \frac{2}{3} = \frac{1}{3}$.

Both cases have a $\frac{1}{2}$ probability, which means the chance of Amy winning is equal to $\frac{1}{2} * \frac{2}{3} + \frac{1}{2} * \frac{1}{3} = \boxed{\frac{1}{2}}$. We note that the answer to part (b) is $\boxed{\frac{2}{3}}$, as determine from Case 1.

Problem 3

Three distinct points are selected randomly among the vertices of a cube.

a) What is the probability that not all three points lie on the same face?

b) What is the probability that the three points can form the vertices of a scalene triangle?

Answer: (a) $\boxed{\frac{4}{7}}$ (b) $\boxed{\frac{3}{14}}$

Solution:
(a) Note that there are $\binom{8}{3} = \frac{8!}{5!3!} = 56$ ways to select the three points on the cube. We perform complementary counting, tallying the number of point triplets that lie on the same face. There are 6 faces, and $\binom{4}{3} = 4$ ways to choose the 3 vertices on each four-vertex face, so a total of $6 * 4 = 24$ ways. The complementary probability is thus $\frac{24}{56} = \frac{3}{7}$, so the probability of not all points being on the same face is $1 - \frac{3}{7} = \boxed{\frac{4}{7}}$.

(b) We note that the only possible distances between vertices on a cube are 1, $\sqrt{2}$, and $\sqrt{3}$. Hence, a scalene triangle must include a cube edge, a face diagonal, and a space diagonal. There are 2 choices for the space diagonal, and 6 choices for a corresponding edge, which must touch a vertex of the space diagonal. Since the face diagonal will be determined by choosing the other two sides, the total number of scalene triangles is $2 * 6 = 12$. Given a denominator of $\binom{8}{3} = 56$, we have a probability of $\frac{12}{56} = \boxed{\frac{3}{14}}$.

Problem 4

A class trip of 10 students is making an evening stop at an Austrian hotel. They have booked five rooms, each of which can house two people. Suppose that people are assigned randomly

to rooms. What is the probability that three students, Peter, Ned, and Flash, are in separate rooms?

Answer: $\boxed{\frac{2}{3}}$

Solution: The total number of ways that 10 students can be divided into 5 distinguishable rooms of 2 is equal to $\frac{10!}{(2!)^5}$, with $(2!)^5$ adjusting for the indifference toward the order of students within each room. We can select 3 rooms for each of Peter, Ned, and Flash in $5 * 4 * 3 = 60$ ways. We then select their roommates from the remaining 7 people, which can transpire in $7 * 6 * 5 = 210$ ways. The 4 unassigned people can be sorted to rooms in $\frac{4!}{(2!)^2} = 6$ ways. Hence, the number of room assignments that separate Peter, Ned, and Flash is equal to $60 * 210 * 6$, so the desired probability is

$$\frac{60 * 210 * 6}{\frac{10!}{(2!)^5}} = \boxed{\frac{2}{3}}.$$

Problem 5

Four teams, the Wizards, Dragons, Jaguars, and Astronauts, are competing for a basketball championship. The Wizard and Dragons each have a 75% chance of defeating the Jaguars or Astronauts, and have a 50% chance against each other. Similarly, the Jaguars and Astronauts have a 50% chance against each other. The teams are randomly assigned to two semifinal matchups, and the winners of each semifinal will play in the finals. What is the probability that the Wizards win the championship?

Answer: $\boxed{\frac{2}{3}}$

Solution: We engineer casework based on the opponent that the Wizards face in the semifinal round.

- *Case 1: The Wizards face the Jaguars.* There is a $\frac{1}{3}$ chance of this assignment and a $\frac{3}{4}$ chance of the Wizards advancing to the finals. The sub-cases depend on which team advances from the other semifinal. (1) The Dragons advance to the finals with $\frac{3}{4}$ probability and the Wizards defeat them in the finals with $\frac{1}{2}$ chance. (2) The Astronauts advance to the finals with $\frac{1}{4}$ probability, and the Wizards win the finals with $\frac{3}{4}$ chance. Hence, the chance of the Wizards winning is

$$\frac{1}{3} * \frac{3}{4} * \left(\frac{3}{4} * \frac{1}{2} + \frac{1}{4} * \frac{3}{4}\right) = \frac{9}{64}.$$

- *Case 2: The Wizards face the Astronauts.* Since the Jaguars and Astronauts are equally matched and swapping them won't change the Wizard's championship chances compared to Case 1, this probability is also simply $\frac{9}{64}$.

- *Case 3: The Wizards face the Dragons.* There is a $\frac{1}{3}$ chance of this assignment and a $\frac{1}{2}$ chance of defeating the Dragons to earn a berth in the finals. We realize that it doesn't

matter who advances from the other final since they are equally matched, meaning that the Wizards will win the finals (if they make it) with $\frac{3}{4}$ chance. The overall probability is $\frac{1}{3} * \frac{1}{2} * \frac{3}{4} = \frac{1}{8}$.

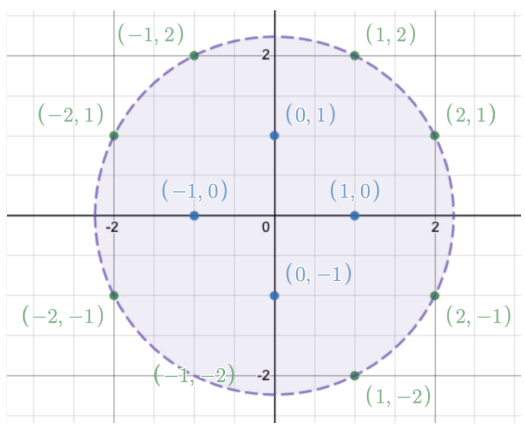
The probability of a Wizards championship is thus $\frac{9}{64} * 2 + \frac{1}{8} = \boxed{\frac{13}{32}}$. This is approximately 41%.

Problem 6

A Mars rover is positioned at $(0, 0)$ in the coordinate plane. For each step, it will move with equal probability either one unit upward, downward, leftward, or rightward. After 7 steps, what is the probability that the Rover is within a distance of $\sqrt{5}$ of the origin?

Answer: $\boxed{\frac{2695}{4096}}$

Solution: We label the movements U (upward), D (downward), L (left), and R (right). We note that to be within $\sqrt{5}$ of the origin, the rover can be at any of the following locations, acknowledging that the sum of the x and y coordinates of the destination point must be odd due to 7 steps which instantly rules out several points.



Owing to symmetry, we can simply compute the probabilities of reaching $(1, 0)$ and $(2, 1)$, which can be each rotated in angular increments around the origin to generate the other points.

- *Case 1:* The robot ends up at $(1, 0)$. There are a few sub-cases: (1) $R = 4$, $L = 3$, $U = D = 0$. The number of arrangements of steps is $\frac{7!}{4!3!0!0!}$. (2) $R = 3$, $L = 2$, $U = D = 1$. We can arrange these steps in $\frac{7!}{3!2!1!1!}$ ways. (3) $R = 2$, $L = 1$, $U = D = 2$. We can arrange these steps in $\frac{7!}{2!1!2!2!}$ ways. (4) $R = 1$, $L = 0$, $U = D = 3$. The quantity of arrangements is $\frac{7!}{1!0!3!3!}$. Adding them up leads to 1225 total arrangements, each of which has a probability of $(\frac{1}{4})^7$, so an overall chance of $\frac{1225}{4^7}$.

- *Case 2:* The robot ends up at $(2, 1)$. There are a few subcases: (1) $R = 4$, $L = 2$, $U = 1$, $D = 0$, which has $\frac{7!}{4!2!1!0!}$ arrangements. (2) $R = 3$, $L = 1$, $U = 2$, $D = 1$, which has $\frac{7!}{3!1!2!1!}$ arrangements. (3) $R = 2, L = 0$, $U = 3, D = 2$, which has $\frac{7!}{2!0!3!2!}$ arrangements. Adding them leads to 735 total arrangements and hence a probability of $\frac{735}{4^7}$.

We note that there are 4 (blue) points under the umbrella of Case 1 due to symmetry, and 8 (green) points for Case 2. Hence, the probability is equal to

$$4 * \frac{1225}{4^7} + 8 * \frac{735}{4^7} = \boxed{\frac{2695}{4096}}.$$

Problem 7

Each person in a cohort of 20 recruits is randomly assigned to one of five factions: Abnegation, Amity, Candor, Dauntless, and Erudite. Each faction must receive at least one new recruit, and no more than 10 new recruits. How many possible distributions of new recruits to factions are possible, assuming that two distributions are considered indistinguishable if they provide the same number of new recruits to each faction?

Answer: $\boxed{\frac{2}{3}}$

Solution: We apply complementary counting, tracing the cases where one faction receives more than 10 new recruits. The total number of cases where the only restriction is at least one new recruit per faction is equal to $\binom{20-1}{5-1} = \binom{19}{4}$ via balls and urns (we arrange 15 recruits, as 5 recruits are preassigned with one per faction, with 4 dividers).

There are 5 factions to choose from that will receive > 10 recruits. We can write $(a + 10) + b + c + d + e = 20$, where a is the number of recruits above 10 allocated to the faction with > 10 recruits and b, c, d, and e are the recruits assigned to the other factions. This means $a + b + c + d + e = 10$ for positive integers, which has $\binom{10-1}{5-1} = \binom{9}{4}$ distributions via balls and urns. Hence, there are $5 * \binom{9}{4}$ distributions in the complementary case.

The number of satisfactory distributions is equal to $\binom{19}{4} - 5 * \binom{9}{4} = \boxed{3246}$.

Problem 8

Each of the positive integer divisors of 15! is written onto a pool ball. All the balls are placed inside a mason jar and mixed. A ball is randomly drawn from the jar.

a) What is the probability that the number written on the ball is odd?

b) What is the probability that the number written on the ball is a perfect square?

Answer: (a) $\boxed{\frac{11}{12}}$ (b) $\boxed{\frac{1}{42}}$

Solution: We can factorize 15! as $2^{11} * 3^6 * 5^3 * 7^2 * 11^1 * 13^1$. We note that a factor of 15! will be of the form $2^a * 3^b * 5^c * 7^d * 11^e * 13^f$ for non-negative exponents, where $a \leq 11$, $b \leq 6$, $c \leq 3$, $d \leq 2$, $e \leq 1$, and $f \leq 1$.

(a) For the factor on the ball to be even, then $a = 0$ since there should be no exponent of 2. Since a has 12 possible choices (0 through 11), the probability of $a = 0$ is $\frac{1}{12}$. This is the complementary probability, so the chance of an odd factor is $1 - \frac{1}{12} = \boxed{\frac{11}{12}}$.

(b) The total number of factors is, via formula, equal to $(11+1)(6+1)(3+1)(2+1)(1+1)(1+1) = 4032$. In order for a factor to be a perfect square, we cannot have 11 and 13 as prime divisors. Additionally, a, b, c, and d must be even, which means we have 6 choices for a, 4 choices for b, 2 choices for c, and 2 choices for d, which accompanies $e = f = 0$. This means the number of perfect square factors is $6 * 4 * 2 * 2 = 96$. The desired probability is $\frac{96}{4032} = \boxed{\frac{1}{42}}$

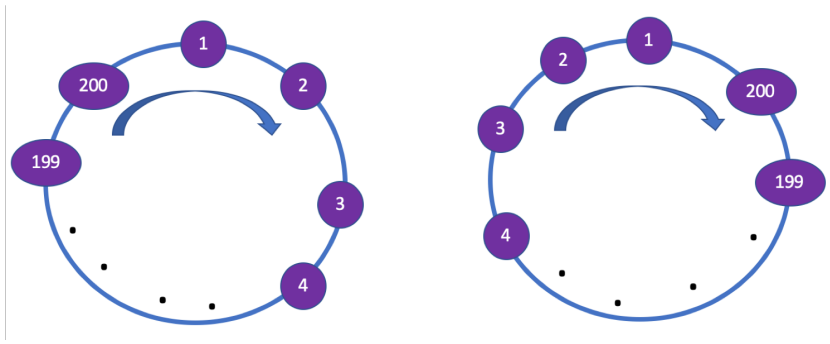
Problem 9

An activity involves 200 people and their phone numbers, which are written onto cards and distributed randomly, with one card provided to each person. The first person starts by calling the number on his card, and the person who receives this call will proceed by calling the number on their card, and so forth. The activity stops when a person receives a call twice. What is the probability that all 200 people will be called during the activity?

Remark: This combinatorial structure is typically called a Hamiltonian cycle.

Answer: $\boxed{\frac{1}{200}}$

Solution: There are 200! total ways that the cards can be distributed to people. In order to for all people to be reached, we must be able to create a cycle of the 200 people such as $1 \to 2 \to 3 \to 4 \to \cdots \to 200$ (this means that person 1 calls person 2, who calls person 3 and so forth), or some permutation of this cycle. This is akin to the distinct number of ways to arrange 200 points on a circle, two of which are illustrated below.



This can be accomplished in $\frac{200!}{200} = 199!$, with the denominator adjusting for rotational symmetry (it doesn't matter where the first point out of 200 is placed). Hence, the desired probability is $\frac{199!}{200!} = \boxed{\frac{1}{200}}$.

Problem 10

Magnus flips n fair coins and Kristoph flips $n+1$ fair coins. What is the probability that Kristoph flips more heads than Magnus?

Answer: $\boxed{\frac{1}{2}}$

Solution: We engineer casework by separating out one coin from Kristoph's $n+1$ coins. We *initially* flip n all of Magnus's coins and n of Kristoph's coins. Let x represent the probability of a tied number of heads initially.

- *Case 1: Magnus has more heads than Kristoph initially.* The flip of Kristoph's $(n+1)$th coin does not matter, since at best, he can only tie the number of heads. There is a 0 chance of Kristoph having more heads here.
- *Case 2: Magnus has the same number of heads as Kristoph initially.* The flip of Kristoph's $(n+1)$th coin must be heads to ensure more heads than Magnus, which occurs with $\frac{1}{2}$ chance.
- *Case 3: Kristoph has more heads than Magnus initially.* The flip of Kristoph's $(n+1)$th coin does not matter, since he will always have more heads. There is a 100% chance of Kristoph having more heads here.

The chance of Case 2 is x, and by symmetry, the chances of Cases 1 and 3 are $\frac{1-x}{2}$ since they have equal probability. Thus, the probability of Kristoph having more heads is equal to

$$\underbrace{\frac{1-x}{2} * 0}_{\text{Case 1}} + \underbrace{x * \frac{1}{2}}_{\text{Case 2}} + \underbrace{\frac{1-x}{2} * 1}_{\text{Case 3}} = \boxed{\frac{1}{2}}.$$

Note this probability is invariant with the exact value of n.

Week 2 Homework

Problem 1

A large airplane contains 300 seats. Alice is assigned to a specific seat. However, due to boarding ticket confusion among the other 299 passengers, each passenger comes into the airplane and sits in a random, unoccupied seat. What is the probability that Alice, who is the last to board, will find her seat unoccupied?

Problem 2

A box contains 5 red marbles and 5 green marbles. Martin removes 7 marbles without replacement from the box. What is the probability that the 7th marble he removes is the final red marble in the box?

Problem 3

Thomas and Teresa are playing a game where they repeatedly flip a fair coin and record the results as a sequence of heads (H) and tails (T). The game will end when either HHH or HTH appears in the sequence. If HHH appears, then Thomas will win; if HTH apperas, then Theresa will win. What is the probability that Teresa wins?

Problem 4

Allison wants to invest 20 million dollars into 4 startups: MagiWater, ICart, Intergen, and Connected. She will only invest money in increments of one million dollars, with each startup receiving anywhere between 1 million and 15 million, inclusive. How many ways can she distribute the 20 million dollars to the startups?

Problem 5

During each step, Gabriel will either take one step forward or one step backward. He is twice as likely to take a forward step than a backward one. If Gabriel takes 6 steps, what is the probability that he is exactly two steps away, either forward or backward, from his original location?

Problem 6

A space army wishes to conquer a row of 6 planets labeled 1 through 6. The army starts at planet 1, and from planet i, the army can use a spaceship to travel to planet $i+1$ or a winged space-plane to travel to planet $i+2$. How many distinct sequences of moves can the army take to move from planet 1 to planet 6? One sample sequence would be to go from planet 1 to 3 using a space-plane, 3 to 4 using a spaceship, and 4 to 6 using a space-plane again.

Week 2 Homework SOLUTIONS

Problem 1

A large airplane contains 300 seats. Alice is assigned to a specific seat. However, due to boarding ticket confusion among the other 299 passengers, each passenger comes into the airplane and sits in a random, unoccupied seat. What is the probability that Alice, who is the last to board, will find her seat unoccupied?

Answer: $\boxed{\frac{1}{300}}$

Solution: This is equivalent to randomly selecting 299 seats from the 300, which means that each seat has a $\frac{1}{300}$ chance of being the unoccupied seat. Thus, the probability that Alice's assigned seat is unoccupied is $\boxed{\frac{1}{300}}$.

Problem 2

A box contains 5 red marbles and 5 green marbles. Martin removes 7 marbles without replacement from the box. What is the probability that the 7th marble he removes is the final red marble in the box?

Answer: $\boxed{\frac{1}{168}}$

Solution: Let R denote a red marble and G denote a green marble. We can write the marbles removed as a sequence such as $GGRRRRR$, which implies a G was removed first and an R last. In order for the 7th marble removed to be the final red, the sequence must have a structure $_R$, where $_$ is a sub-sequence of 4 Rs and 2 Gs. The number of possible sequences is thus $\binom{6}{2} = 15$. The probability of any sequence of 5 Rs and 2 Gs is equal to

$$\frac{5*4*3*2*1*2*1}{10*9*8*7*6*5*4} = \frac{1}{2520}.$$

Hence, the desired probability is $15 * \frac{1}{2520} = \boxed{\frac{1}{168}}$.

Problem 3

Thomas and Teresa are playing a game where they repeatedly flip a fair coin and record the results as a sequence of heads (H) and tails (T). The game will end when either HHH or HTH appears in the sequence. If HHH appears, then Thomas will win; if HTH apperas, then Theresa will win. What is the probability that Teresa wins?

Answer: $\boxed{\frac{3}{5}}$

Solution: Since both Thomas and Teresa require a heads to win, we will start our analysis from the first heads (H) that is flipped. Let p be the probability that Thomas wins here. We build casework based on the next flip.

- *Case 1: The next flip is H.* Thomas will win with another H ($\frac{1}{2}$ chance) as $HH\mathbf{H}$ will occur. If the next flip is T, then Thomas needs a T to prevent the sequence $HH\mathbf{TH}$ from occurring. As we have $HH\mathbf{TT}$ ($(\frac{1}{2})^2$ chance), the game effectively resets as both Thomas and Teresa will need to wait for the next H to occur. Since the probability of Case 1 is equal to $\frac{1}{2}$, then Thomas's chances of winning with this case is equal to $\frac{1}{2}\left[\frac{1}{2} + (\frac{1}{2})^2 * p\right]$.
- *Case 1: The next flip is T.* Thomas will need another T to prevent an immediate loss ($\frac{1}{2}$ chance), leading to $HT\mathbf{T}$ and the game effectively resetting for the same reasons as above. Since the probability of Case 2 is equal to $\frac{1}{2}$, the chance of Thomas winning with Case 2 is equal to $\frac{1}{2} * \frac{1}{2} * p$.

Thomas's probability of winning is thus $\frac{1}{2}\left[\frac{1}{2} + (\frac{1}{2})^2 * p\right] + \frac{1}{2} * \frac{1}{2} * p$, which is equal to p by definition. Thus,

$$\begin{aligned}\frac{1}{2}\left[\frac{1}{2} + (\frac{1}{2})^2 * p\right] + \frac{1}{2} * \frac{1}{2} * p &= p \\ \frac{3}{8}p + \frac{1}{4} &= p \\ \frac{1}{4} &= \frac{5}{8}p\end{aligned}$$

This means that $p = \frac{2}{5}$, so the probability that Teresa wins is equal to $1 - \frac{2}{5} = \boxed{\frac{3}{5}}$. It may make intuitive sense that Teresa has an advantage as HTH seems more likely to be somewhere in a long sequence than three consecutive heads.

Hint: It may be helpful to design an event tree for these type of problems.

Problem 4

Allison wants to invest 20 million dollars into 4 startups: MagiWater, ICart, Intergen, and Connected. She will only invest money in increments of one million dollars, with each startup receiving anywhere between 1 million and 15 million, inclusive. How many ways can she distribute the 20 million dollars to the startups?

Answer: $\boxed{953}$

Solution: We essentially want to find the number of positive integer solutions to $x_1 + x_2 + x_3 + x_4 = 20$, where x_1, x_2, x_3, and x_4 are the number of millions of dollars invested in MagiWater, ICart, Integen, and Connected, respectively. The number of solutions is $\binom{20-1}{4-1} = \binom{19}{3} = 969$ by *balls and urns*, as we essentially have $4 - 1 = 3$ dividers and $20 - 4 = 16$ freely depositable millions (1 million is required per company).

We employ complimentary counting, counting the number of ways to distribute money where one company receives > 15 million. Without the loss of generality, suppose that company 1 receives the most and let x_1^* denote the number of millions above 15. Thus,

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 = (x_1^* + 15) + x_2 + x_3 + x_4 &= 20 \\ x_1^* + x_2 + x_3 + x_4 &= 5 \end{aligned}$$

The number of solutions here is $\frac{5-1}{4-1} = 4$ by *balls and urns*. Since there were 4 companies to choose to receive the most, the total number of solutions is $4 * 4 = 16$. Since there were 969 total, the total ways to distribute money where *no company receives* > 15 *million* is equal to $969 - 16 = \boxed{953}$.

Problem 5

During each step, Gabriel will either take one step forward or one step backward. He is twice as likely to take a forward step than a backward one. If Gabriel takes 6 steps, what is the probability that he is exactly two steps away, either forward or backward, from his original location?

Answer: $\boxed{\frac{15}{32}}$

Solution: Denote a forward step as F and a backward step as B. We partition our problem into two cases:

- *Case 1: Gabriel is* 2 *steps forward from his original location.* This means that we have 4 Fs and 2 Bs. The probability of this is equal to $\binom{6}{2} * (\frac{1}{2})^6$, as the number of ways to arrange 4 Fs and 2 Bs in a sequence is $\binom{6}{2}$ and each sequence of 6 binary moves has a $(\frac{1}{2})^6$ chance of occurring. Hence, the probability is $\frac{15}{64}$.
- *Case 2: Gabriel is* 2 *steps backward from his original location.* Since Gabriel chooses a forward or backward step with equal probability, this case is symmetrical to Case 1, just that Gabriel ends up backward from his original location. The probability is also $\frac{15}{64}$.

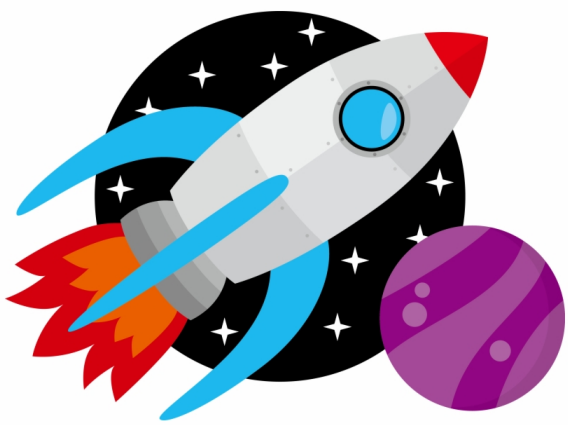
The total probability is equal to $\frac{15}{64} * 2 = \boxed{\frac{15}{32}}$, which is approximately 47%.

Problem 6

A space army wishes to conquer a row of 6 planets labeled 1 through 6. The army starts at planet 1, and from planet i, the army can use a spaceship to travel to planet $i+1$ or a winged space-plane to travel to planet $i+2$. How many distinct sequences of moves can the army take to move from planet 1 to planet 6? One sample sequence would be to go from planet 1 to 3 using a space-plane, 3 to 4 using a spaceship, and 4 to 6 using a space-plane again.

Answer: $\boxed{8}$

Solution: Let f_n be the number of ways that the army can reach planet n from planet 1. Evidently, $f_1 = 1$ (just staying still) and $f_2 = 1$ (use a spaceship to travel to planet $i+1$). Note that if an army has reached planet n, it must have either came from planet $n-1$ using a spaceship or from planet $n-2$ using a winged space-plane. Hence, the number of ways to reach planet n is the sum of the number of ways to reach planet $n-1$ and the number of ways to reach planet $n-2$. More succinctly,

$$f_n = f_{n-1} + f_{n-2}.$$

As $f_1 = 1$ and $f_2 = 1$, we can find $f_3 = 1+1 = 2$, $f_4 = 2+1 = 3$, $f_5 = 3+2 = 5$, and $f_6 = 5+3 = \boxed{8}$. As $n = 8$ is small here, it is not particularly difficult to directly count f_8, but recursion provides a more general blueprint for computing the answer.